Constrained Delaunay tetrahedrisation refinement

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Outline

Introduction
- Who am I?
- Constrained Delaunay tetrahedrisation
- State of the art

The refinement algorithm
- General principle
- Edge collapse
- A new operation

Results
- Some graphics
- Comparison to TetGen

Conclusion

Quick presentation

- Max Royer

Figure: A ragu I made in Genova

Quick presentation

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- ENS of Lyon

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Quick presentation

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- End of first year of master

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Quick presentation

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- ENS of Lyon
- End of first year of master
- Internship for 3 months

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Let's add some constraints

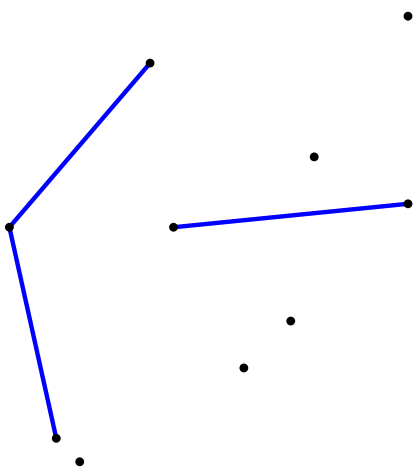

Figure: Edge constraints for Delaunay triangulation

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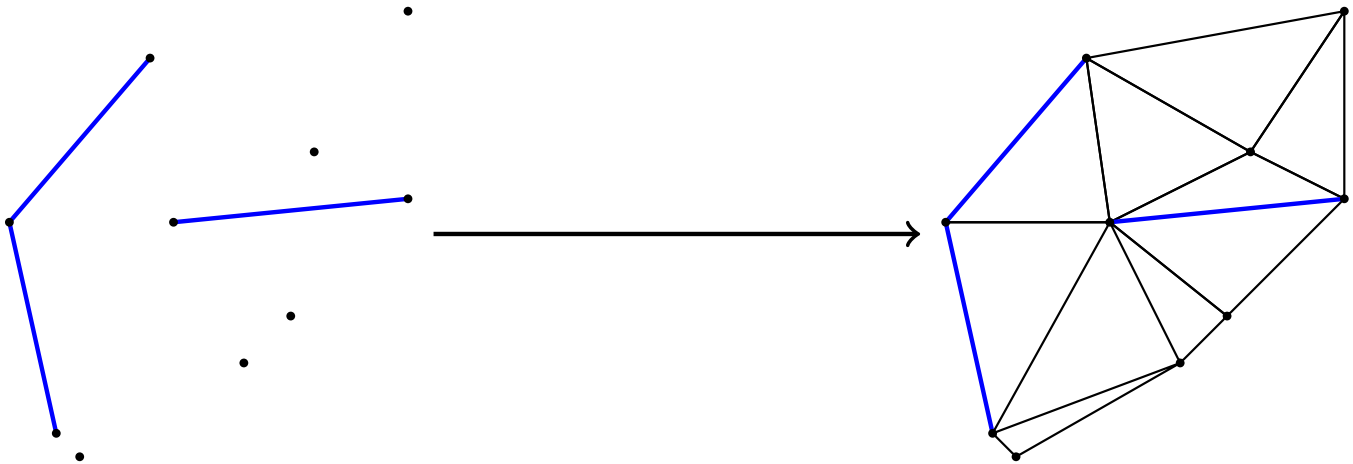

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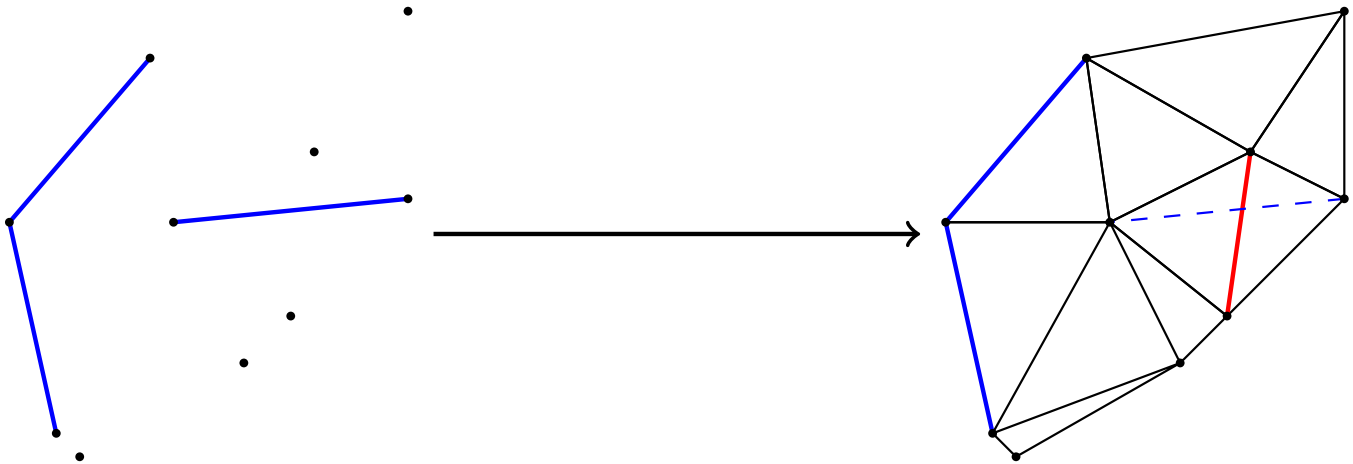

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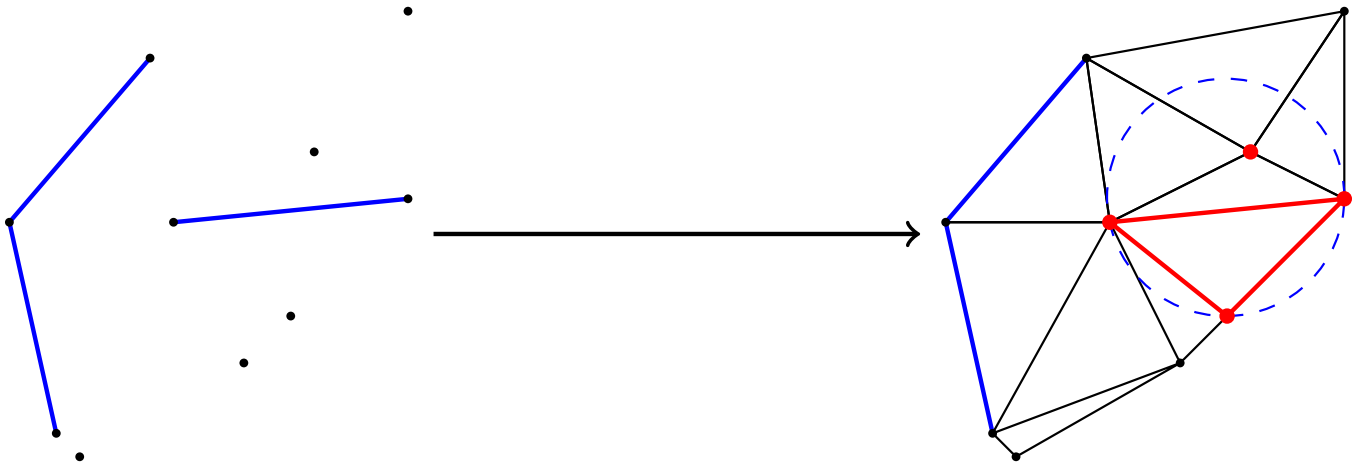

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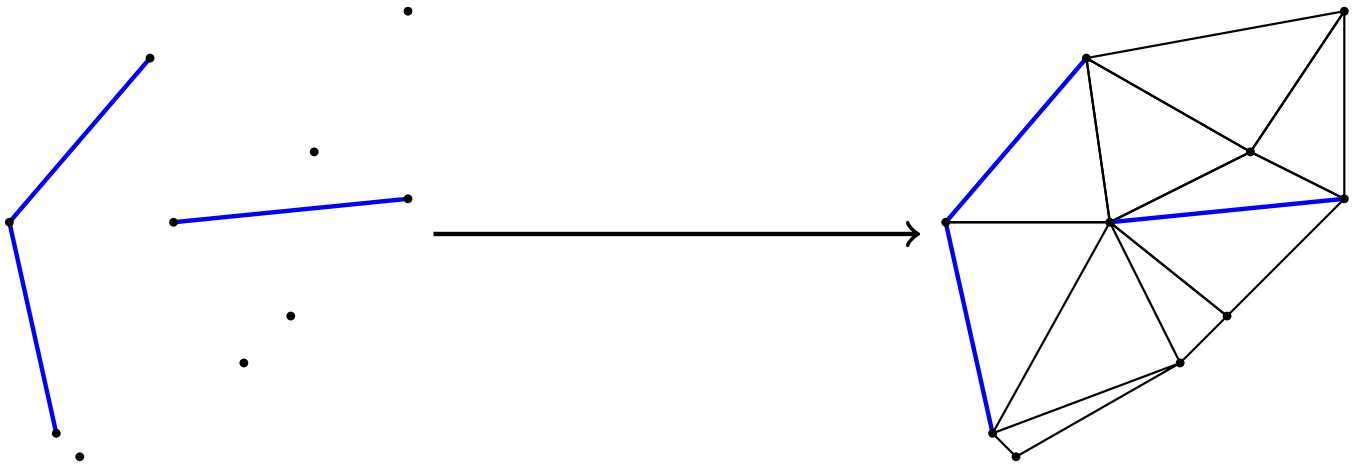

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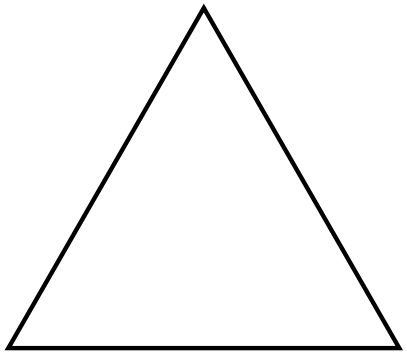
An edge is either a constraint or is locally Delaunay

Differences from 2D to 3D?



From this ...

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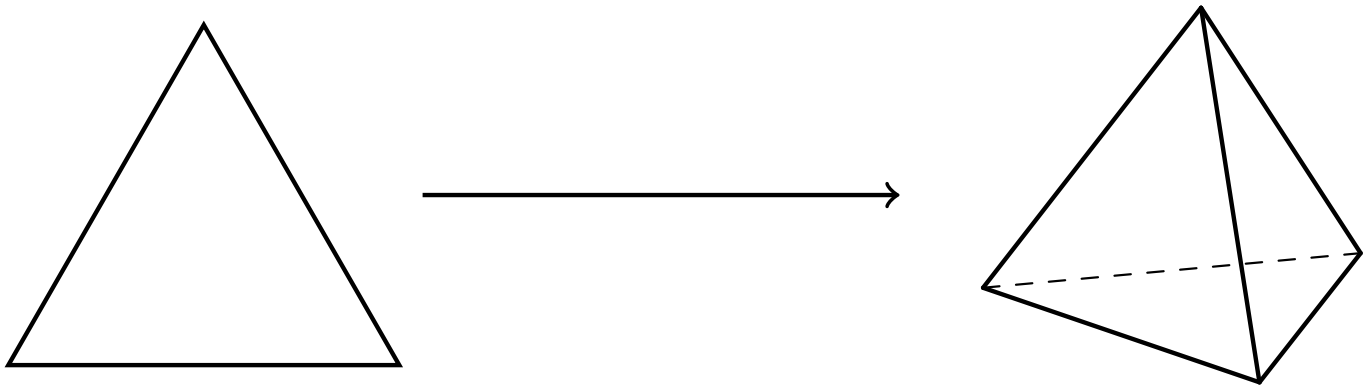

From this ... ... to this

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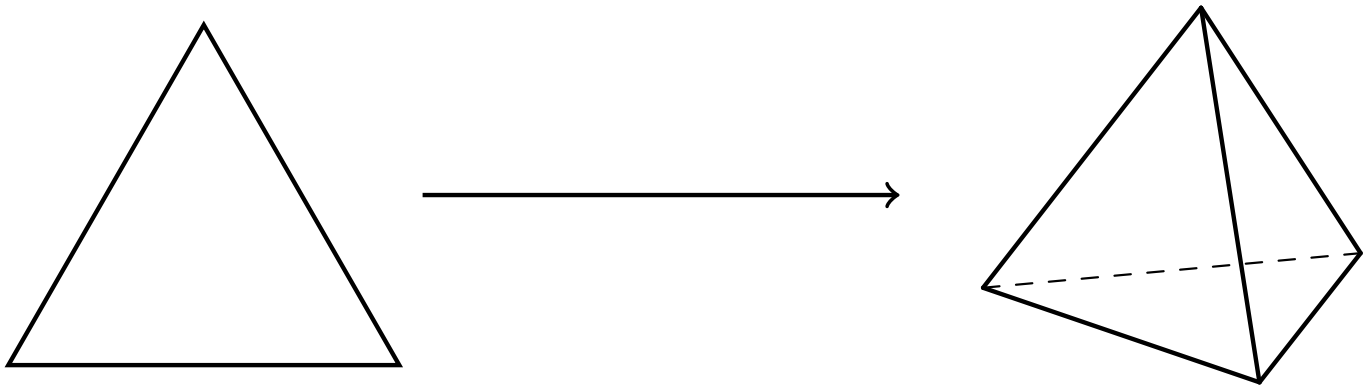

From this ... ... to this

Only ??

Not really...

... it gets harder

Not really...

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- Slivers appear

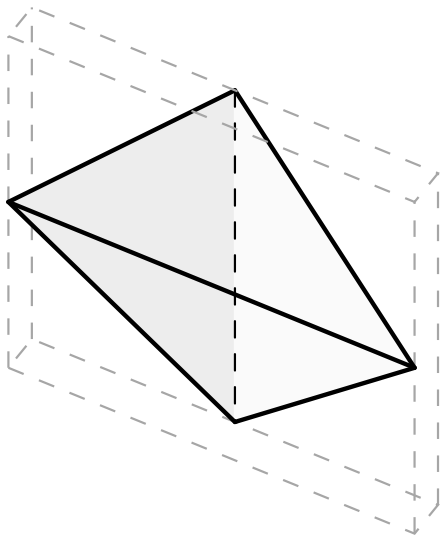

Figure: Example of sliver

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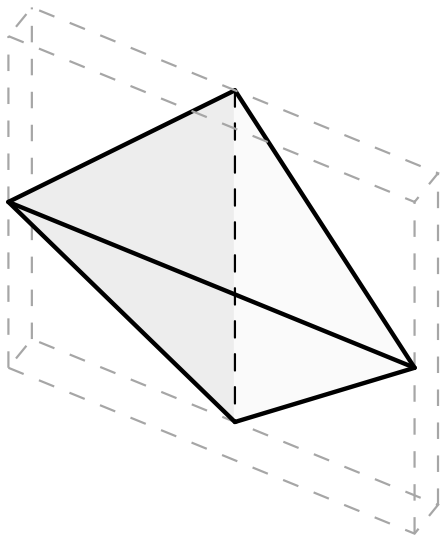

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- Constrained Delaunay tetrahedrisation are not always possible

TetGen [Si, 2015]

Already exisiting software

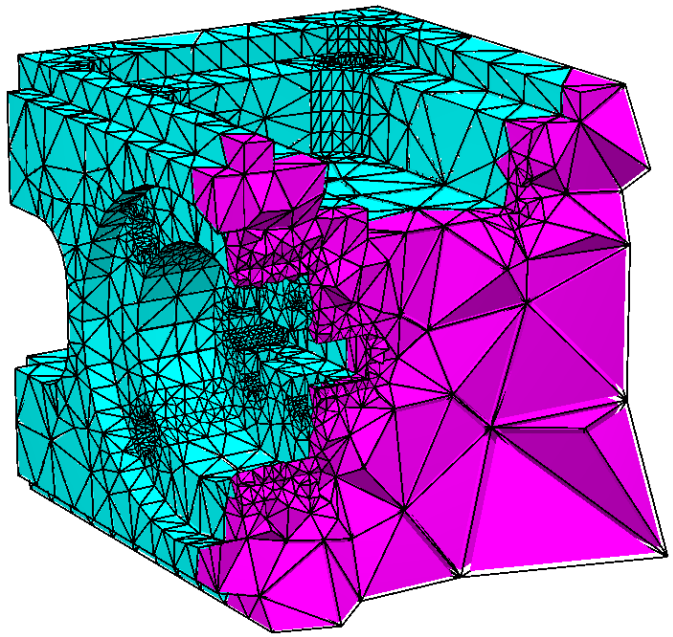
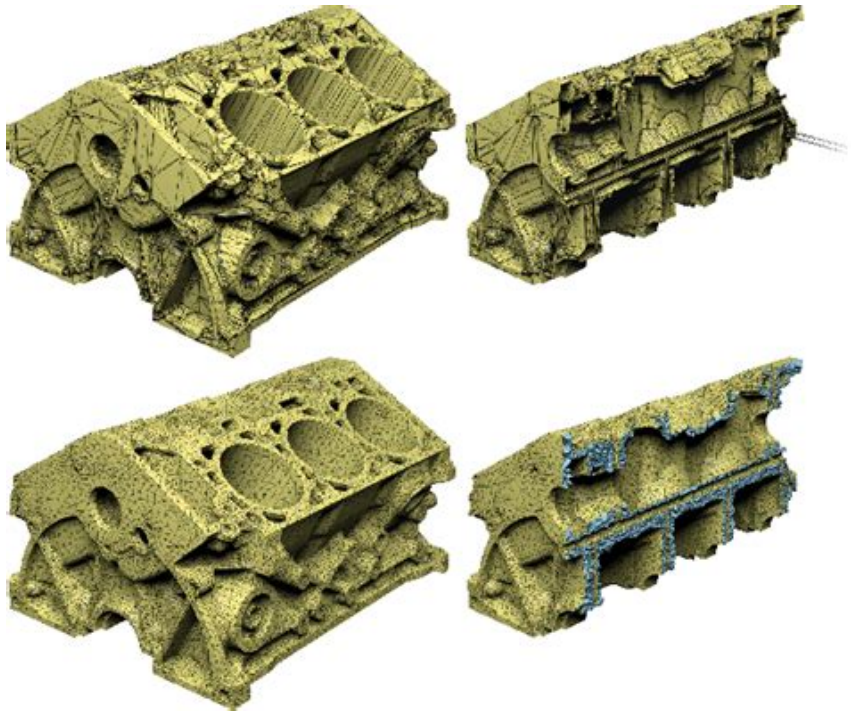


Figure: A model tetrahedrized by TetGen

- Not so robust, crashs on some models

Tetrahedral meshing in the wild (Tetwild) [Hu, 2018]



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- Not strict on boundary

Quality measure

How to say if a tetrahedron is *"good"*?

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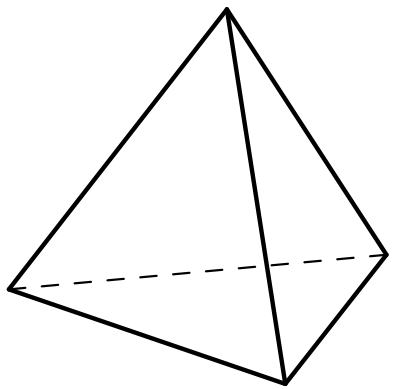

Minimized on a regular tetrahedron ($= 3$)

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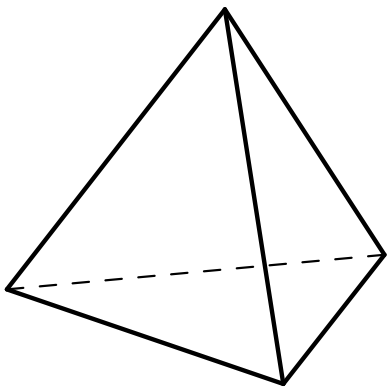

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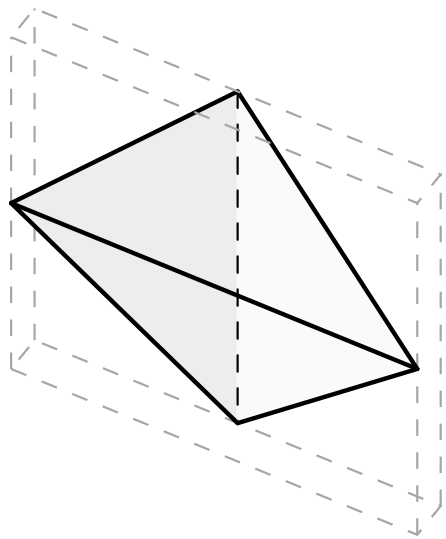
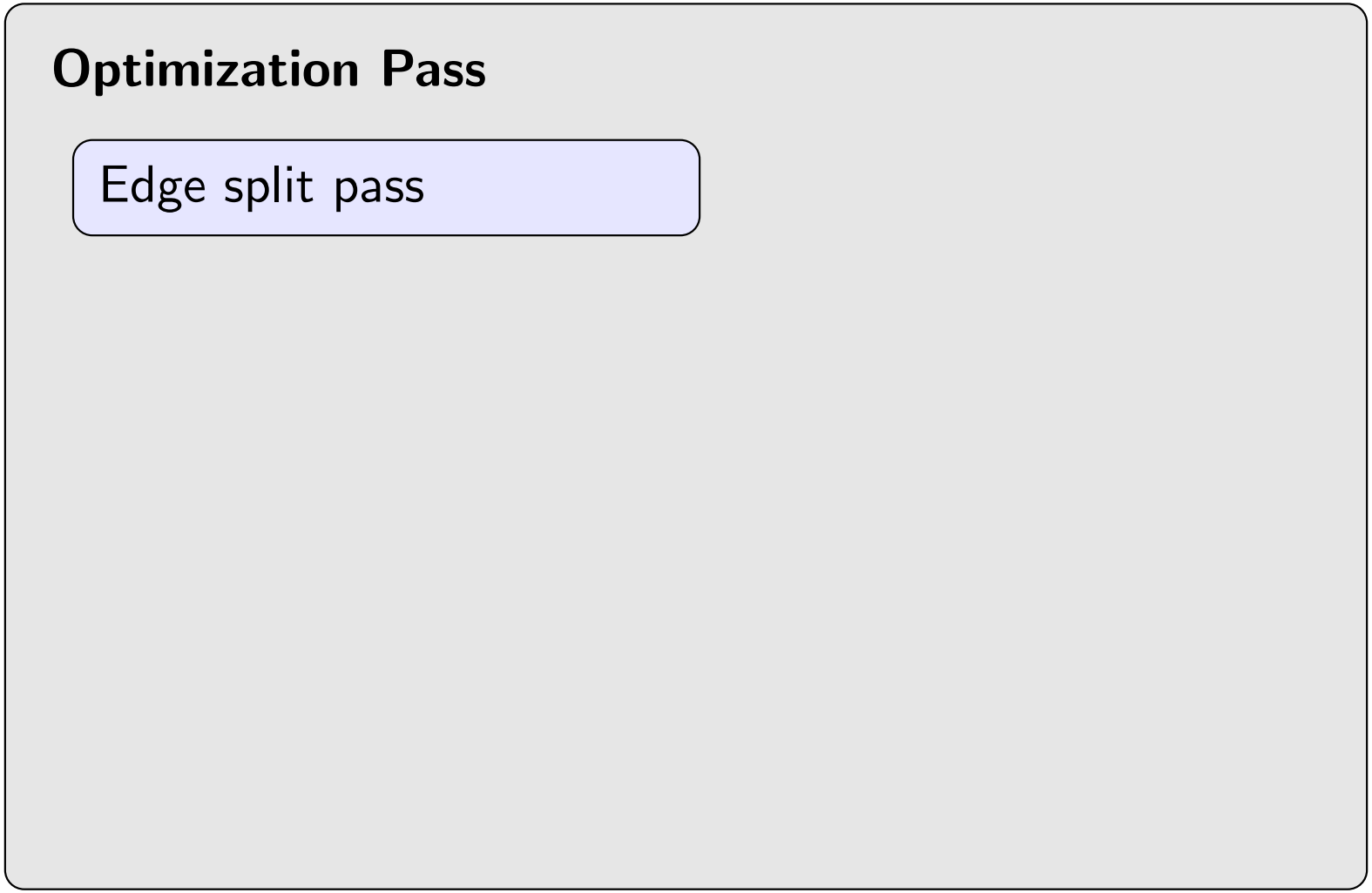


Big value on bad tetrahedron ($\sim 10^3 - 10^4$)

Optimization Pass

Figure: Optimization process scheme



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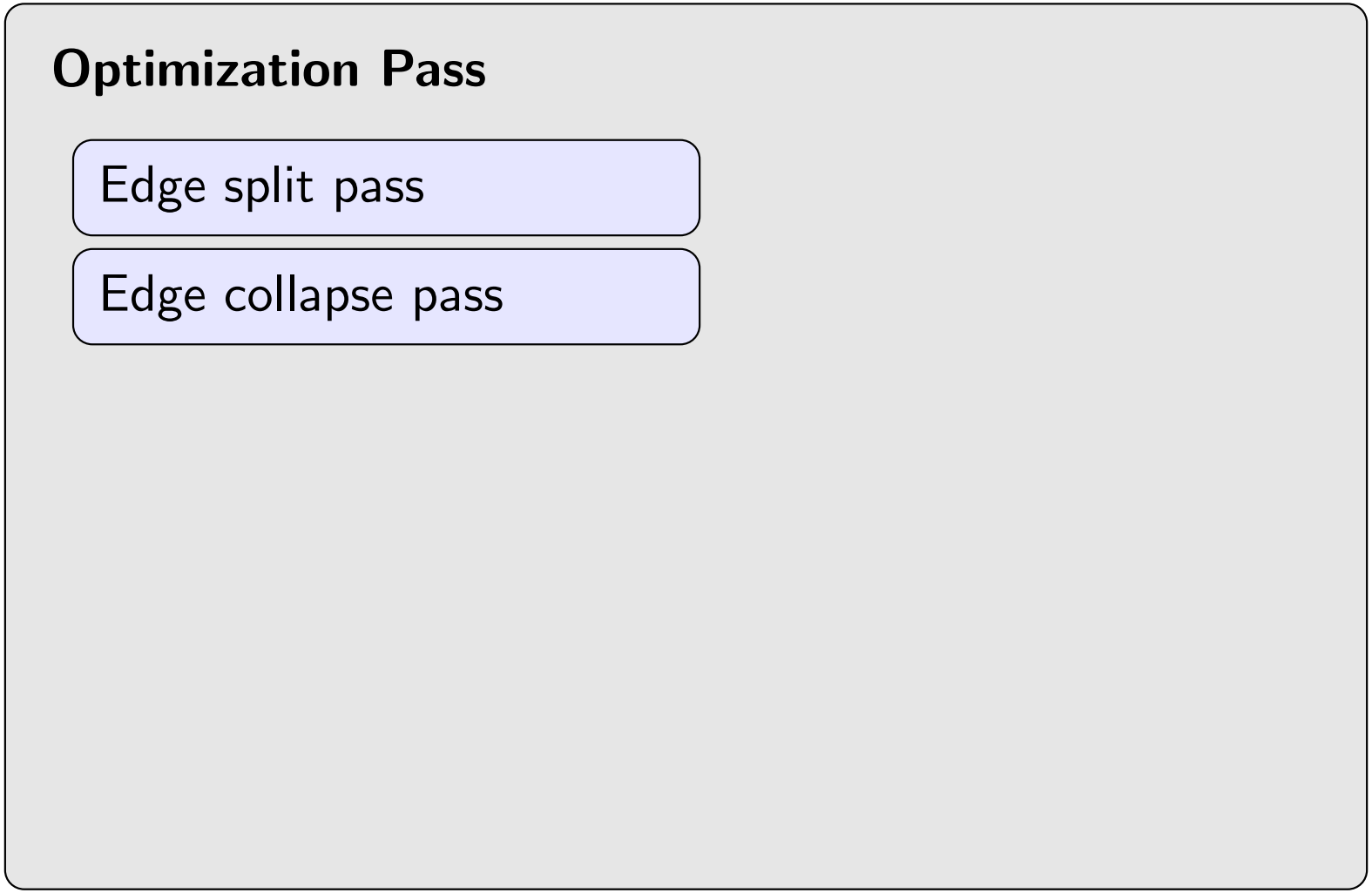

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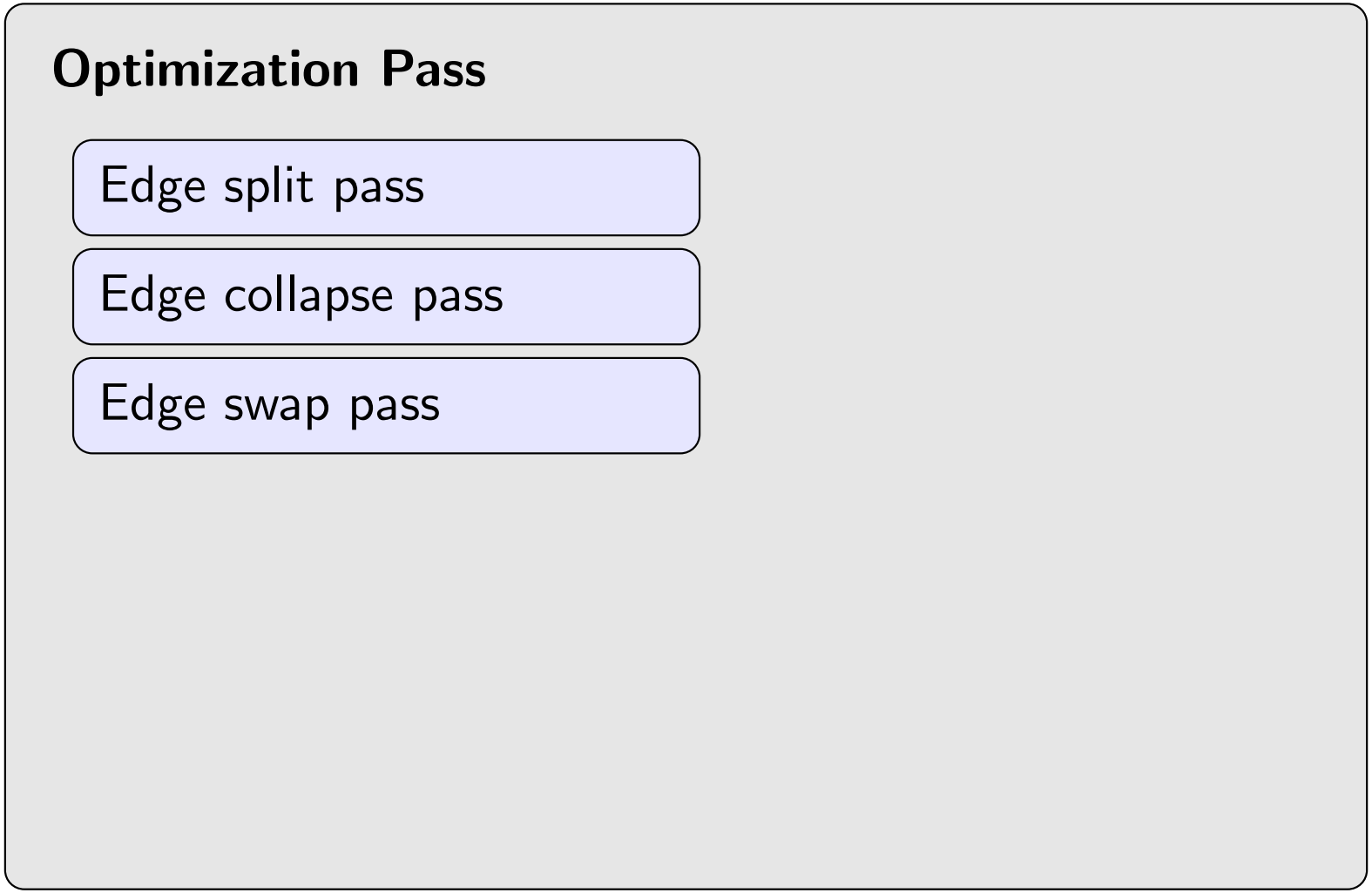

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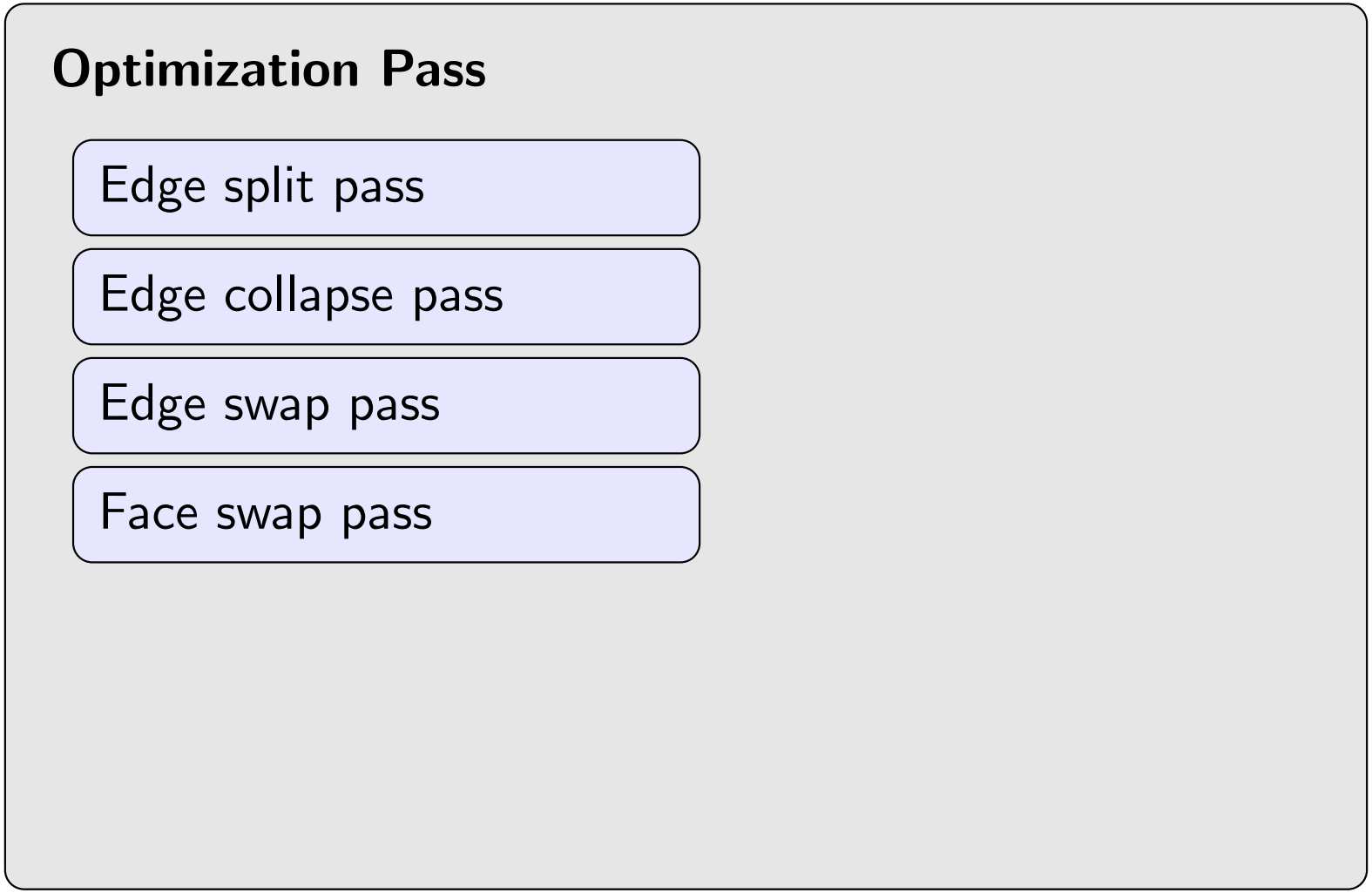

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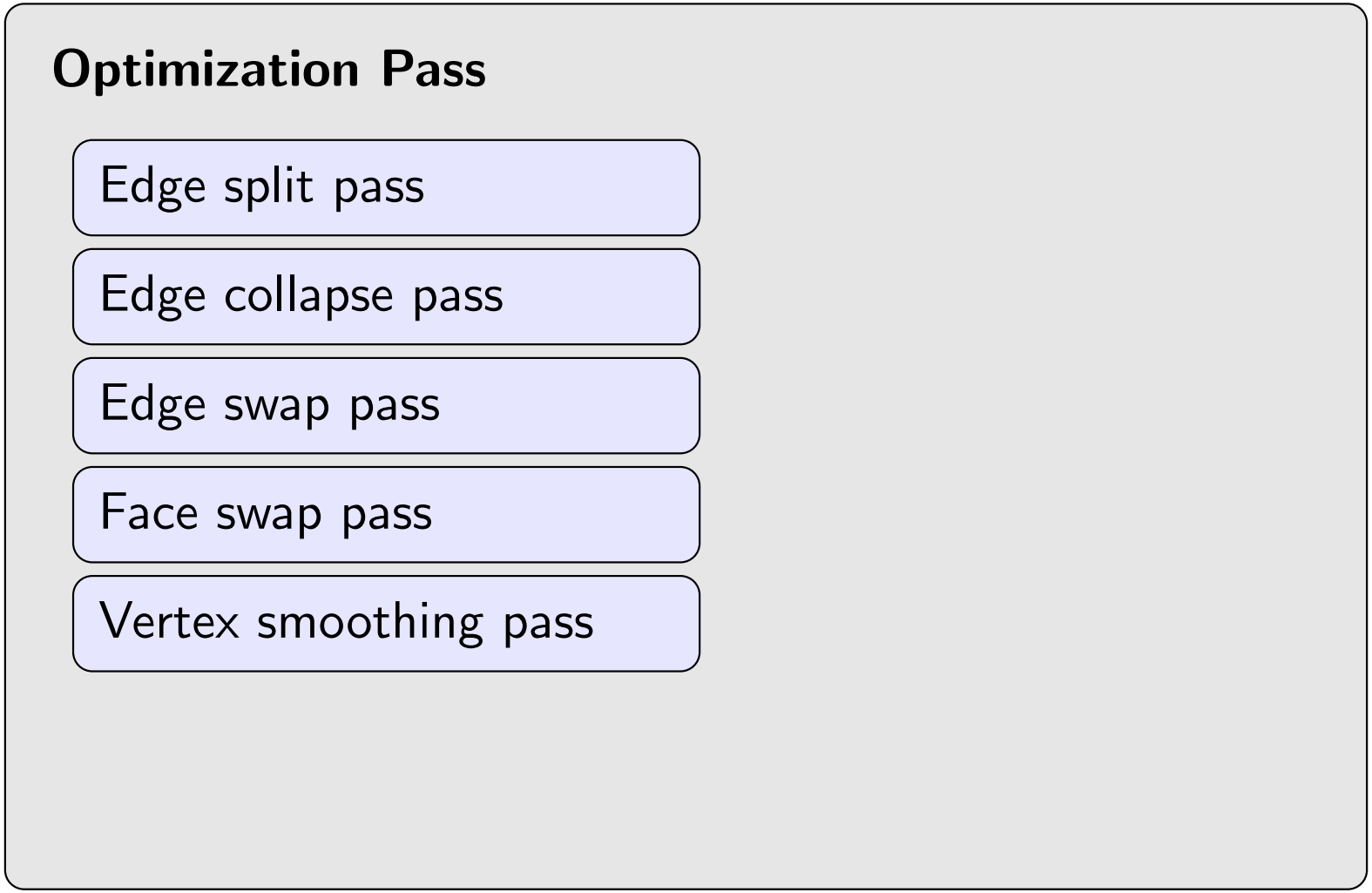

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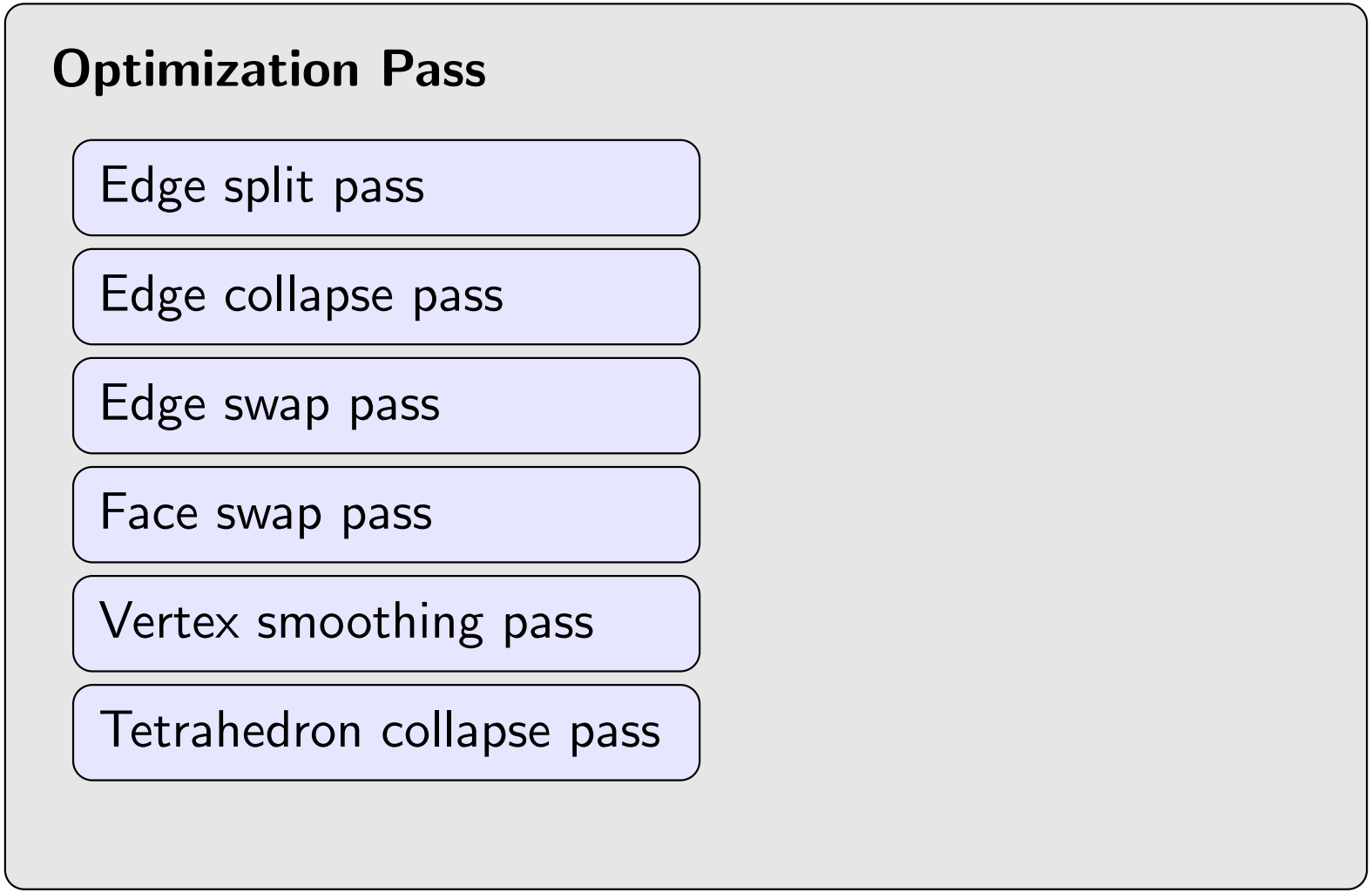

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Local operation pass

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Local operation pass

Simulation pass

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Simulation pass $\longrightarrow$ Yields a queue Q

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Applying pass, while Q not empty:

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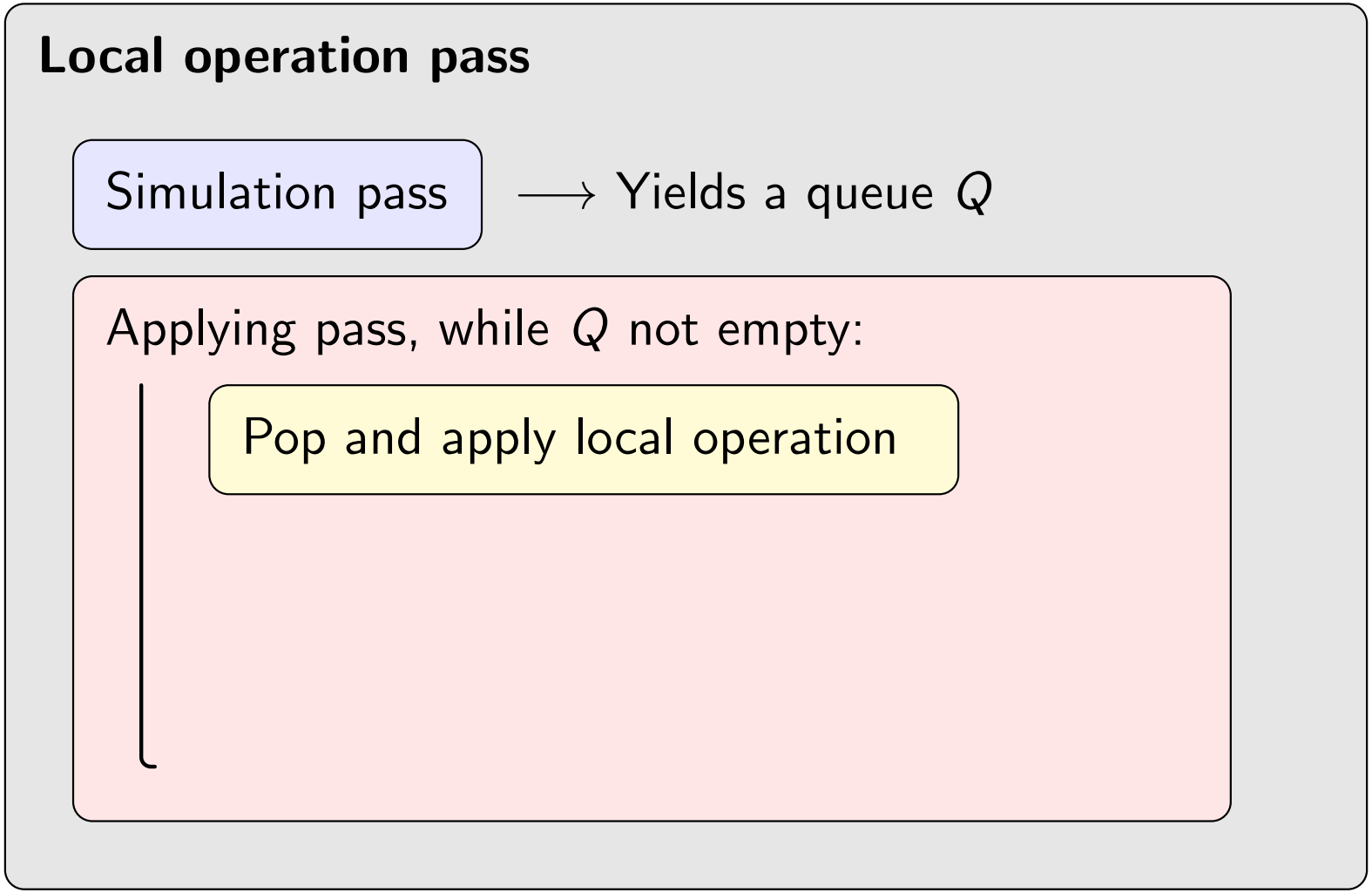

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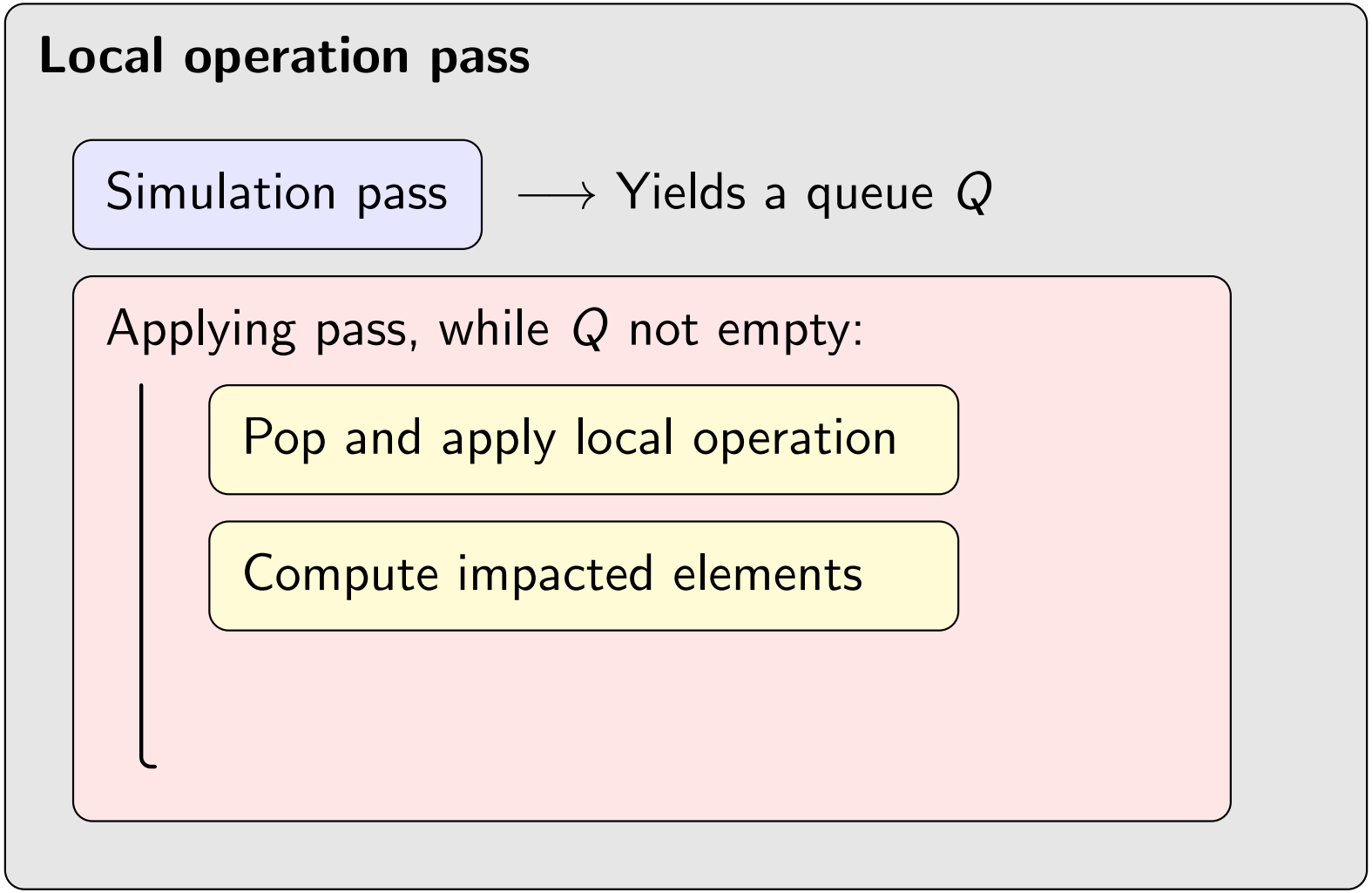

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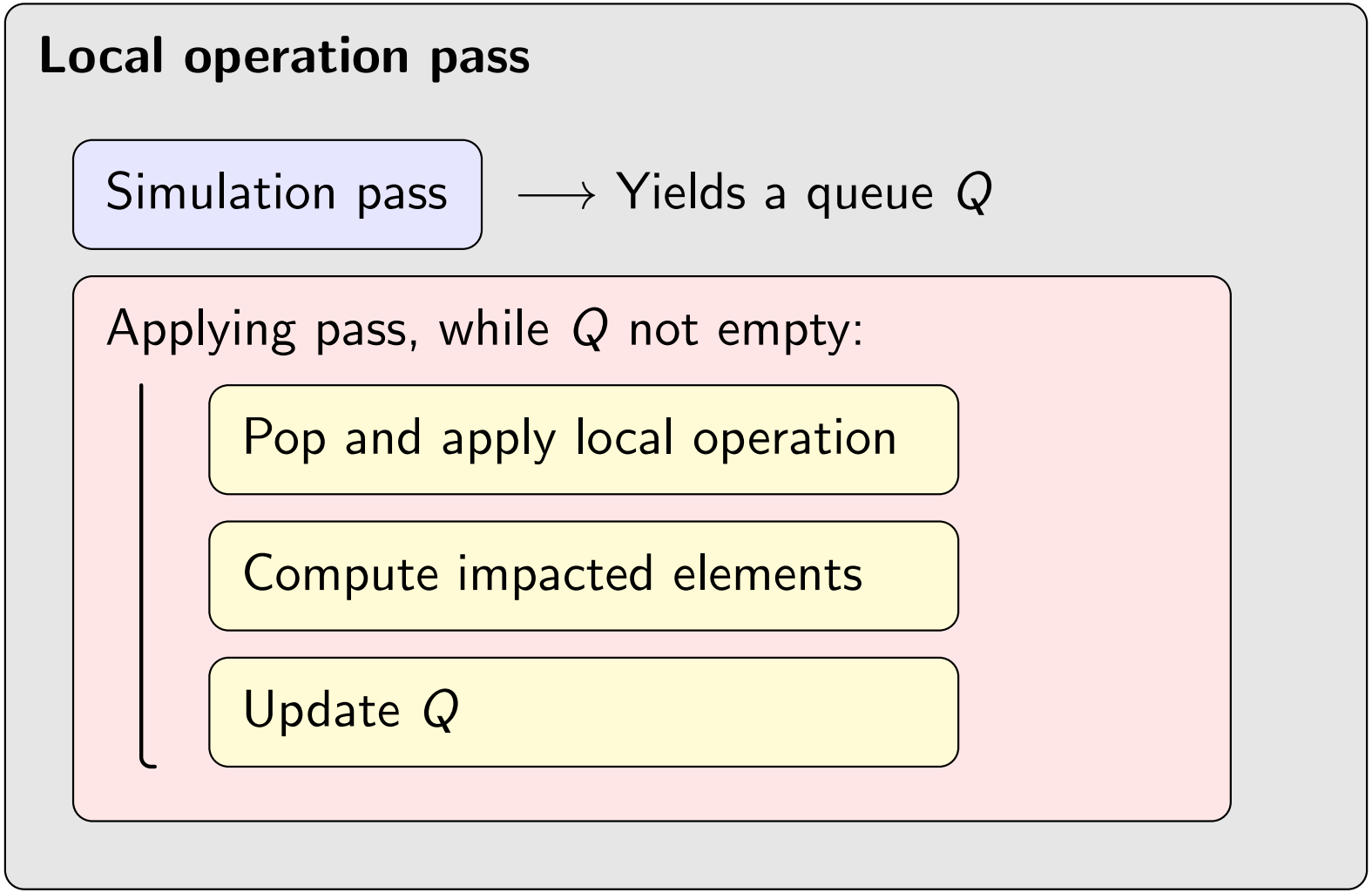

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Overview

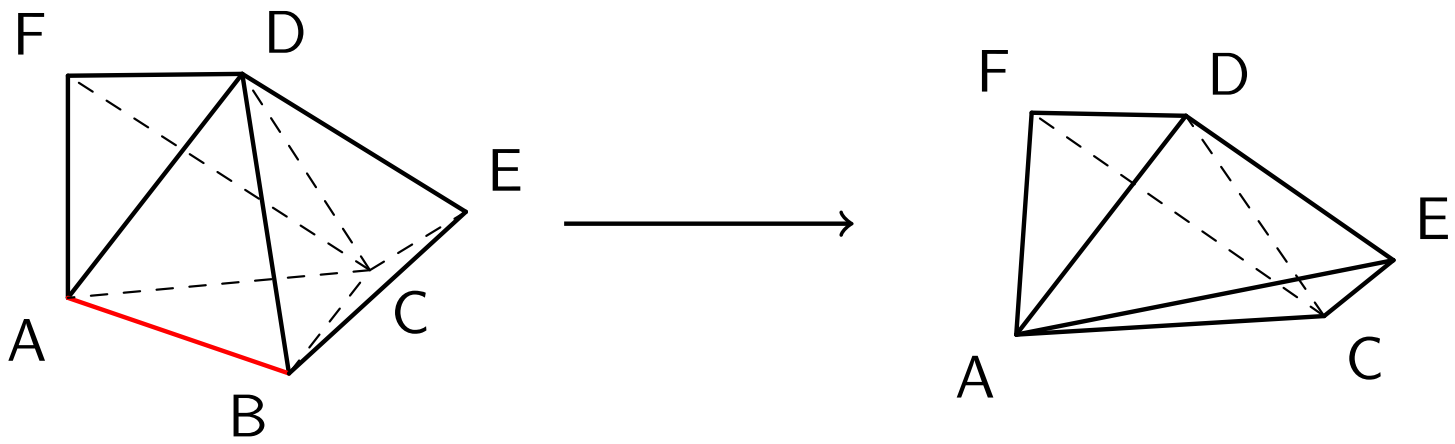

Figure: Example of collapsing an edge

Simulation

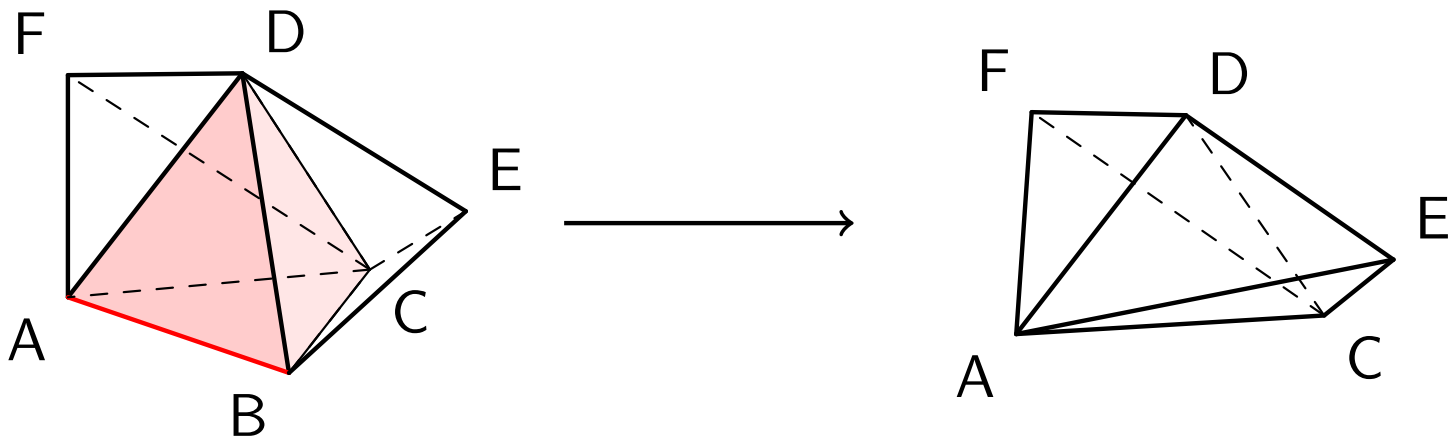

Figure: Impacted tetrahedra by an edge collapse

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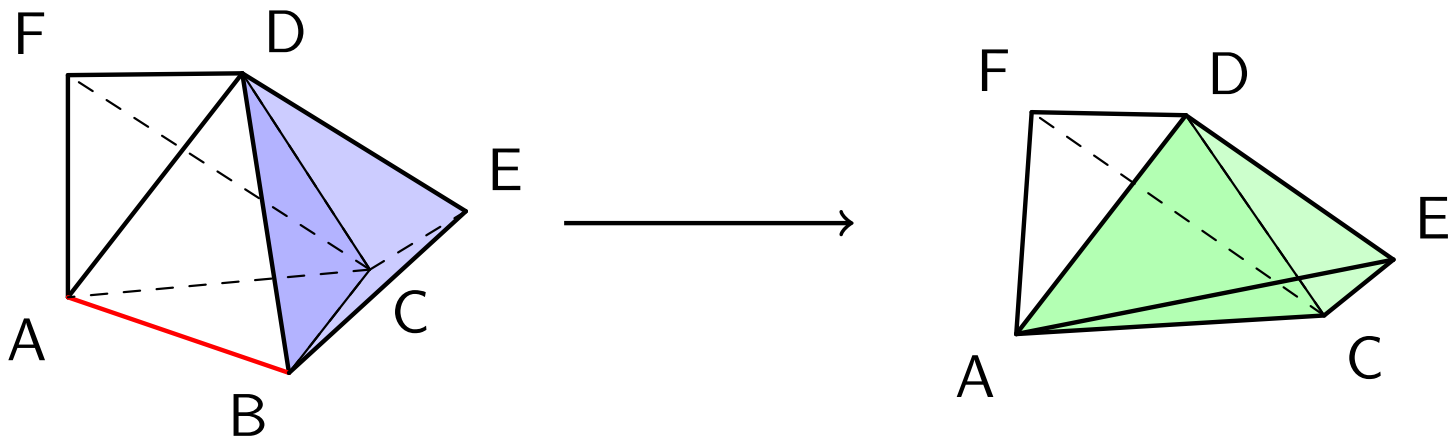

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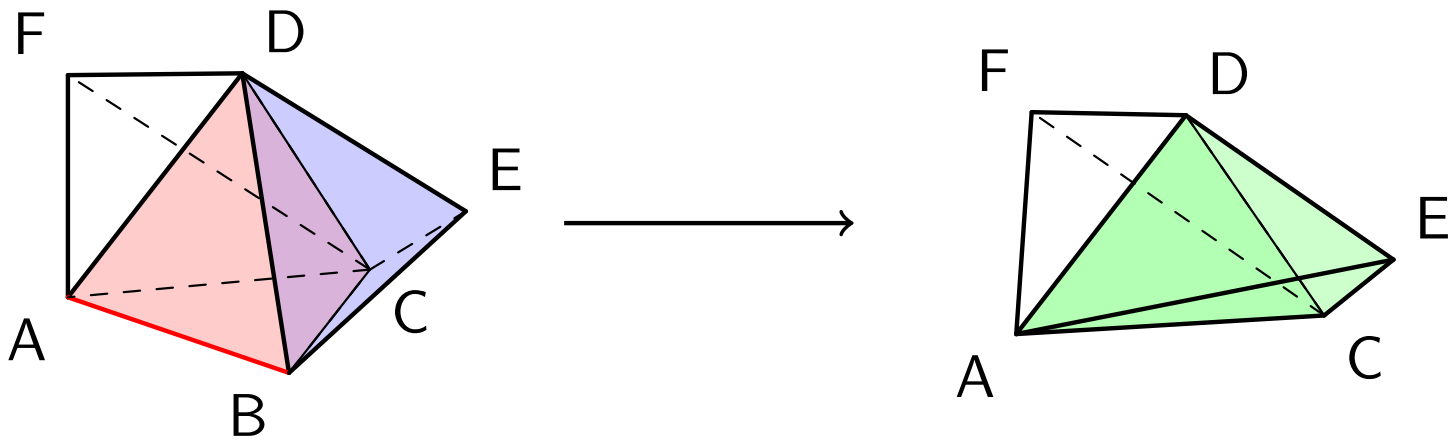

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Quality before:

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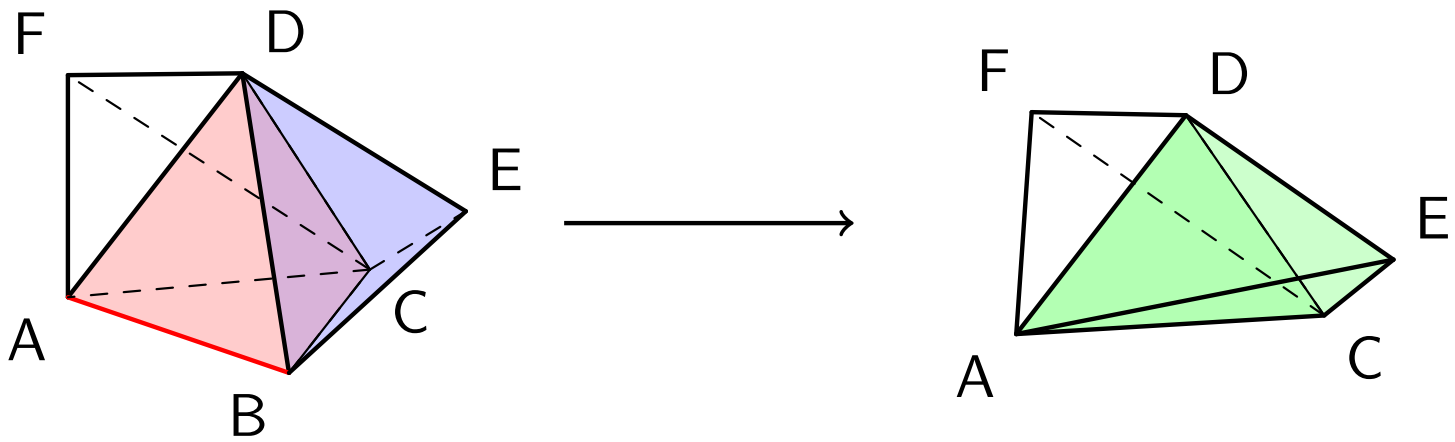

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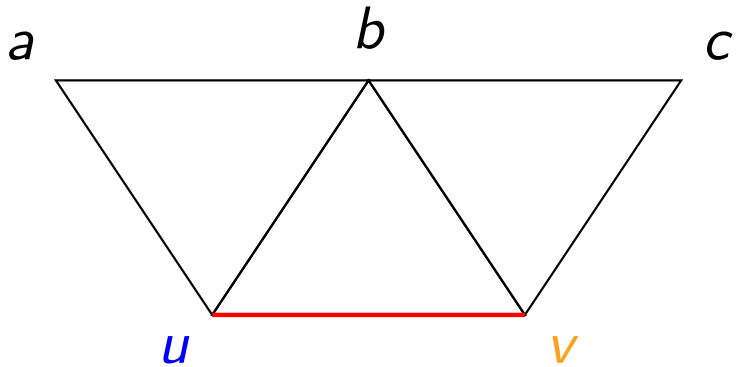

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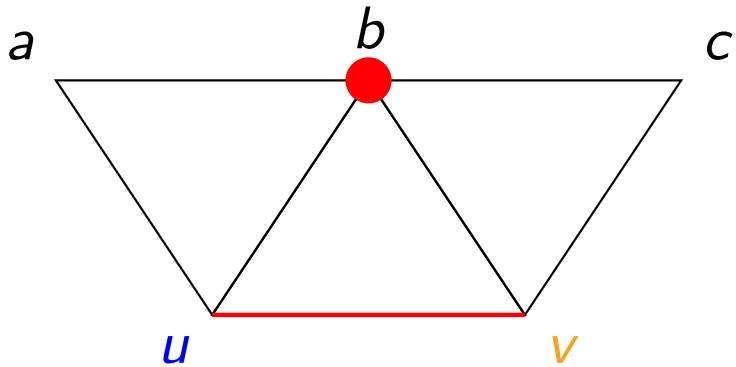

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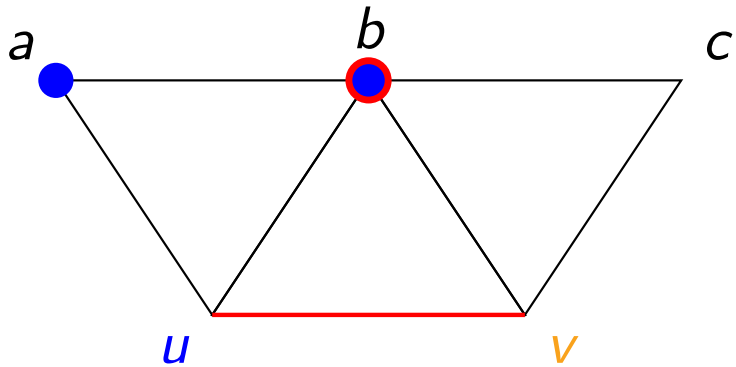

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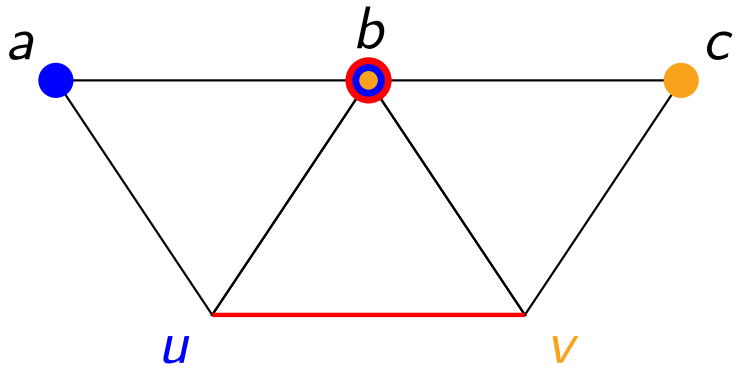

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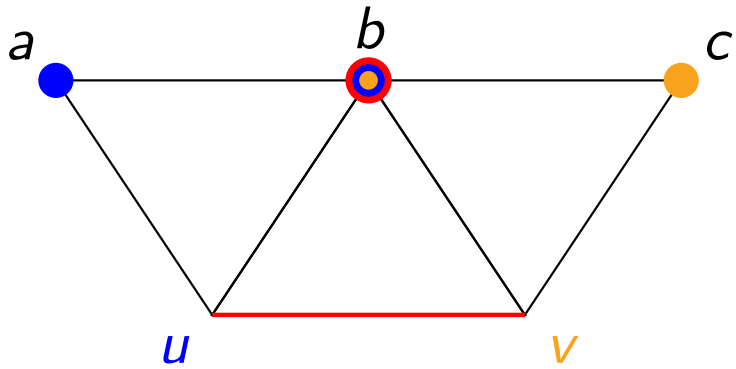

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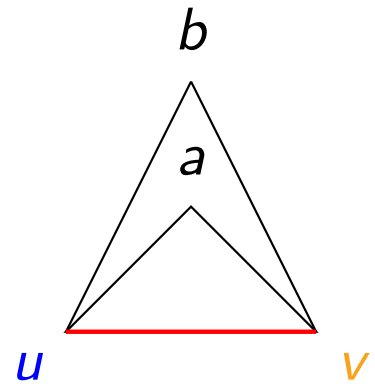

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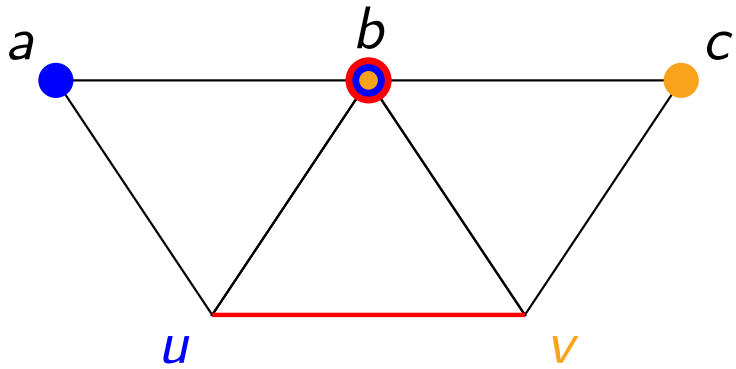

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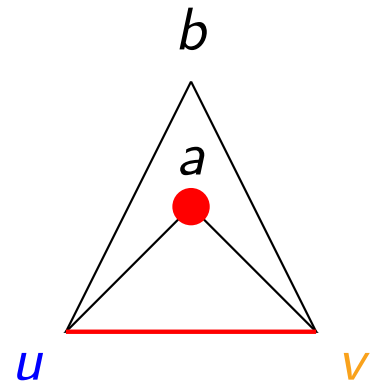

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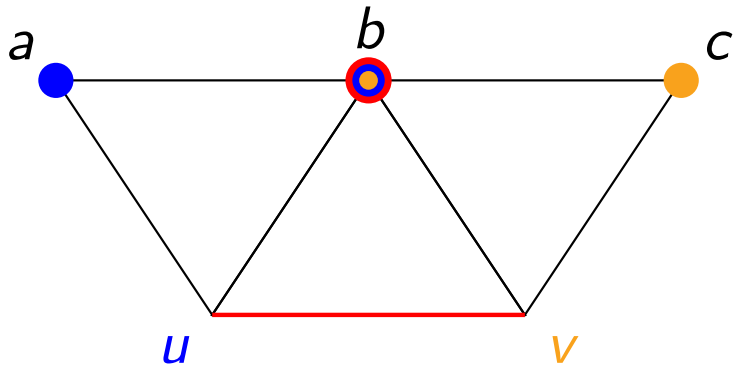

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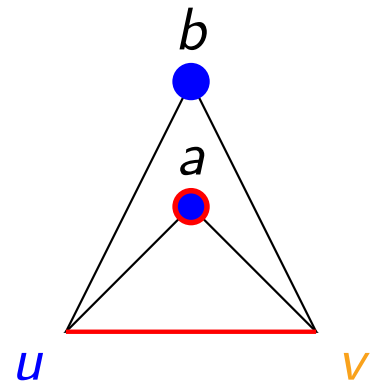

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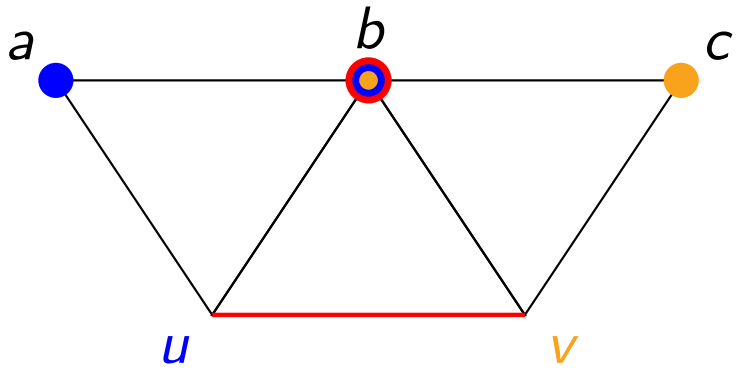

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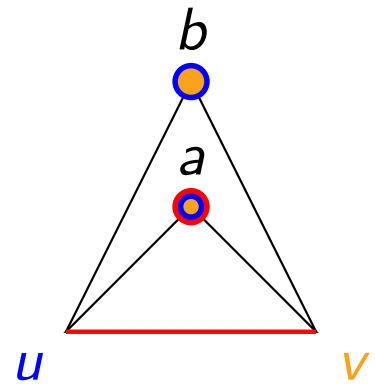

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Get rid of slivers

Some slivers resisted known operation $\Rightarrow$ Create a new operation

Figure: Tetrahedron collapsing process

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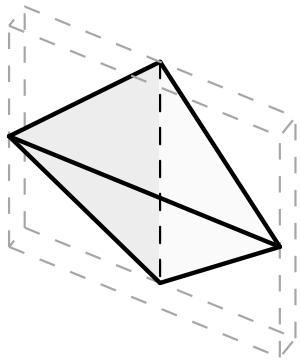

Figure: Tetrahedron collapsing process

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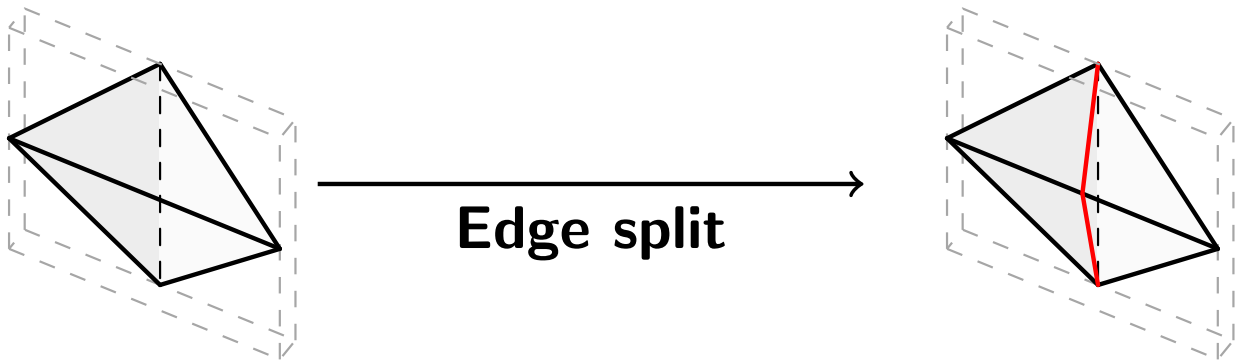

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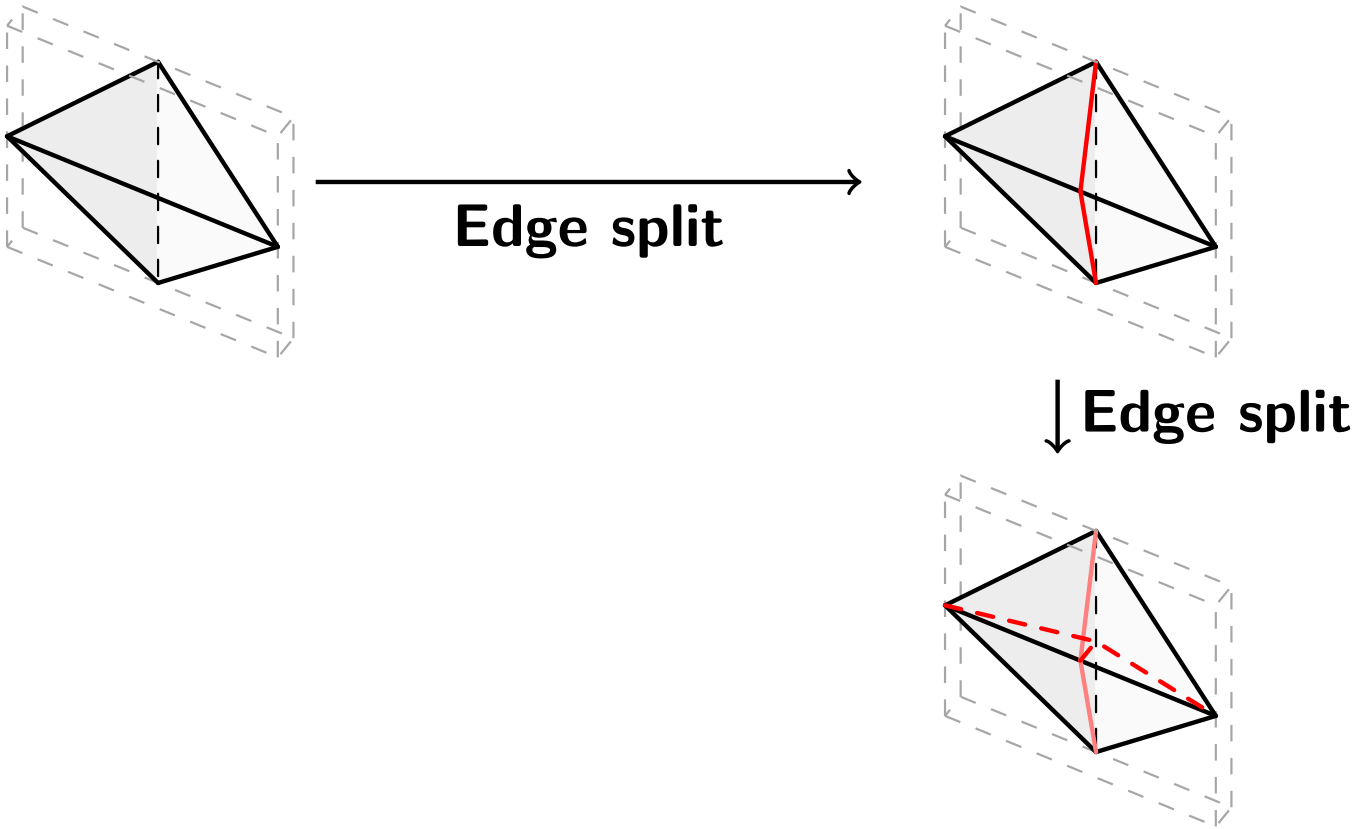

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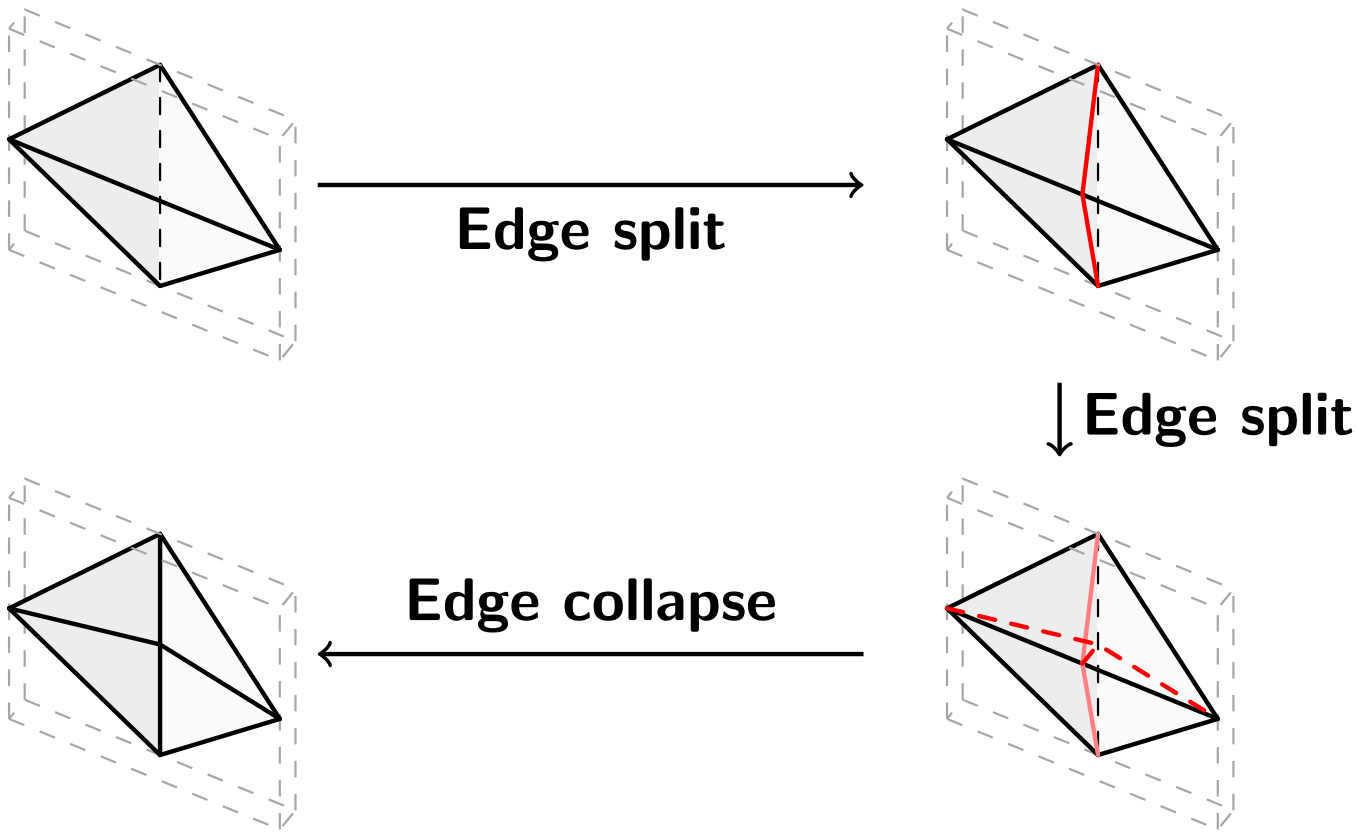

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An example

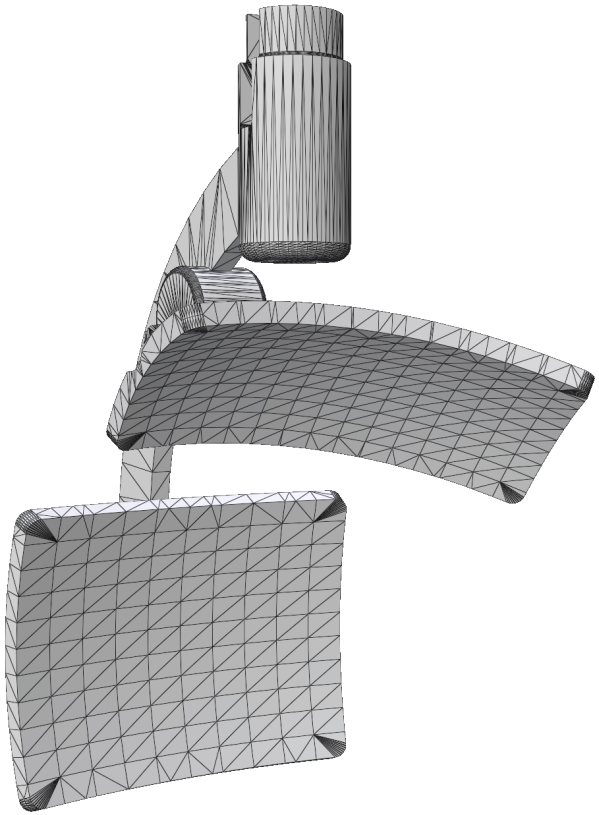
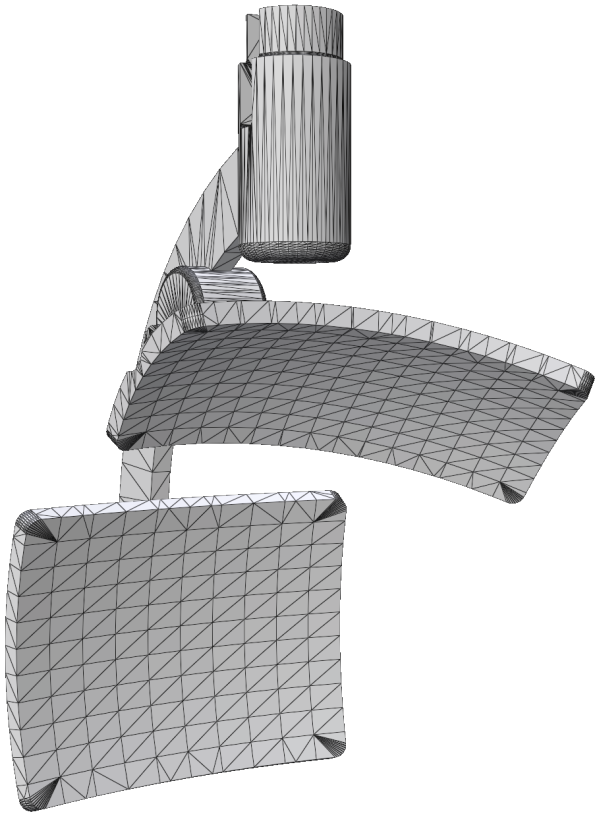
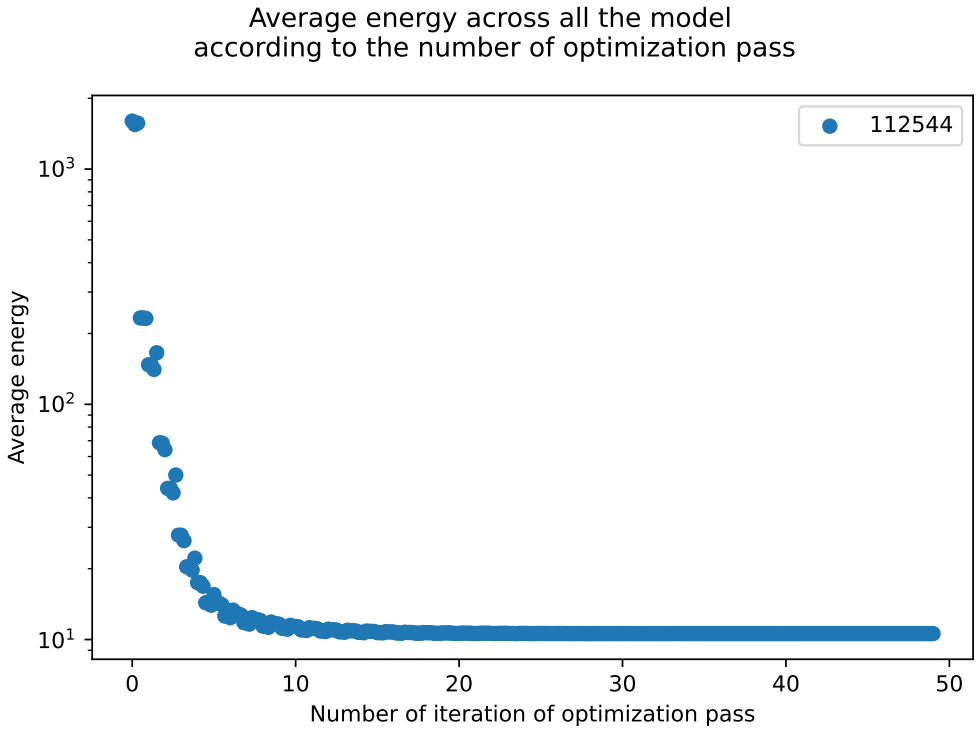


Figure: The model 112544

An example





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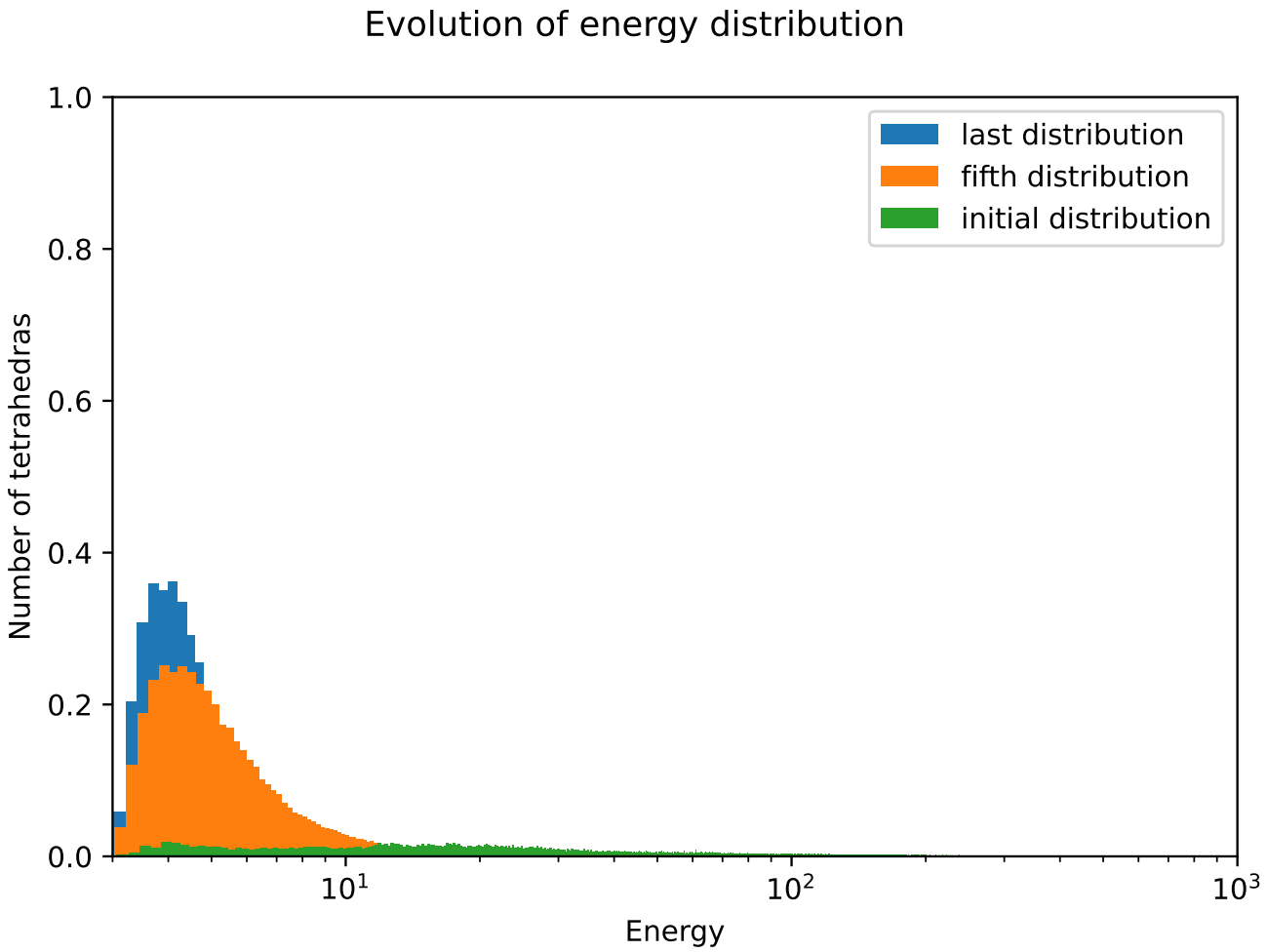
Evolution of the distribution



Comparison

- \+ More robust

	CDT	TetGen
Fails	0%	25,2%

Table: Stats on the smallest 500 models of *Thingi10k*

Comparison

- \+ More robust
- \- Slower

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- Lack of time

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Future work and improvements

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- Make some operations finer
  - Split vertex not always at midpoint

References

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